

AI System Architecture and Microservices

Overview:

The OmniBrain AI system is organized into decoupled microservices communicating via asynchronous event queuing.

The core services include the Document Parser Service, the Embedding Generation Service, the Vector Storage Engine, and the LLM Orchestrator.

Microservices Communication:

1. The Document Parser Service ingests raw files (PDFs, Markdown, text) and extracts structured textual data into chunks.
2. The Embedding Generation Service converts text chunks into 384-dimensional dense vectors using transformer-based models.
3. The Vector Storage Engine utilizes FAISS (Facebook AI Similarity Search) for sub-millisecond similarity queries.
4. The LLM Orchestrator manages prompt construction, context grounding, and API interactions with Google Gemini.

Retrieval Accuracy and Evaluation Metrics

Evaluation Framework:

To guarantee grounded responses, the AI module measures retrieval precision using three key metrics:

- Mean Reciprocal Rank (MRR): Evaluates where the first relevant document chunk appears in the ranked search results.
- Hit Rate at K (Hit@K): Measures whether at least one relevant document chunk is present in the top-K retrieved results.
- Context Precision: Measures the ratio of relevant chunks in the retrieved context to total chunks passed to the LLM.

Similarity Threshold Calibration:

For cosine similarity under the all-MiniLM-L6-v2 model:

- Scores ≥ 0.55 indicate High Confidence relevant matches.
- Scores between 0.35 and 0.55 indicate Moderate Relevance context.
- Scores < 0.35 are treated as low-confidence noise and filtered out before LLM generation.